
*RESTORATION AND ENHANCEMENT OF DEGRADED IMAGES USING DIGITAL
IMAGE PROCESSING*

DIP Mini Project Report

Submitted by:

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RESTORATION AND ENHANCEMENT OF DEGRADED IMAGES USING DIGITAL IMAGE PROCESSING

1. INTRODUCTION

Digital images are widely used in photography, communication, documentation, surveillance, healthcare, and many other applications. However, the quality of an image can decrease because of noise, blur, poor acquisition conditions, or transmission-related disturbances.

Image restoration is an important area of Digital Image Processing that aims to recover an image from a degraded version. The objective is to reduce unwanted degradation while preserving important visual information such as edges, structures, and details.

In this project, four images are used as reference images. Gaussian noise is artificially added to these images to simulate image degradation. Three processing approaches are then studied:

- Median Filtering
- Gaussian Filtering
- CLAHE-based contrast enhancement

The results are compared with the original images using both visual observation and objective image-quality metrics.

The project therefore provides a practical demonstration of how Digital Image Processing techniques can be used to improve the quality of degraded images.

2. PROBLEM STATEMENT

Images can become degraded due to noise and other disturbances, resulting in reduced visual quality and loss of useful information.

The problem addressed in this project is:

How can a degraded image containing noise be restored using simple Digital Image Processing techniques while maintaining similarity to the original image?

To study this problem, four images are used as reference images. Gaussian noise is introduced artificially to create degraded versions. Median Filtering and Gaussian Filtering are then applied for noise reduction, followed by CLAHE-based contrast enhancement.

The restored outputs are compared with the original images using MSE, PSNR, and SSIM.

3. OBJECTIVES

The main objectives of the project are:

1. To understand the effect of Gaussian noise on digital images.
2. To simulate image degradation by adding Gaussian noise to four input images.
3. To restore degraded images using Median Filtering.
4. To restore degraded images using Gaussian Filtering.
5. To improve the local contrast of restored images using CLAHE.
6. To visually compare the original, degraded, filtered, and enhanced images.
7. To evaluate the restoration techniques using MSE, PSNR, and SSIM.
8. To compare the performance of the different techniques and identify the most effective method for the selected images.

4. IMAGE DATA DESCRIPTION

4.1 Image Selection

Four images were used for the experiment. The images were uploaded directly into the Google Colab environment and used as reference images.

The selected images contain normal visual content and were used to represent clean input images before artificial degradation.

Number of images

4 images

Image format

The images were uploaded in common image formats such as JPG/PNG.

Image preprocessing

Before processing, each image was:

1. Read using OpenCV.

2. Converted from BGR format to RGB format.
3. Resized to **512 × 512 pixels**.

Resizing was performed so that all four images had the same dimensions during processing and evaluation.

4.2 Original Images

The four original images used in the project are shown below.



Figure 1: Original images used for the experiment

4.3 Dataset Limitation

The experiment uses only four images. Therefore, the results are intended to demonstrate the behavior of the selected restoration techniques rather than represent the performance of the methods on a large image dataset.

In addition, the images were not originally degraded photographs. Instead, Gaussian noise was artificially introduced to simulate image degradation. This provides a controlled environment for comparing restoration techniques.

5. METHODOLOGY

The complete workflow of the project is:

Original Image → Gaussian Noise Addition → Degraded Image → Restoration → Contrast Enhancement → Evaluation → Comparison

The methodology consists of the following stages.

5.1 Input Image

Four images are used as clean reference images.

The images are loaded into Google Colab using Python and OpenCV.

5.2 Image Preprocessing

The uploaded images are converted from BGR to RGB format and resized to 512×512 pixels.

This provides a consistent image size for all subsequent operations.

5.3 Image Degradation

To simulate a degraded photograph, Gaussian noise is added to each image.

Gaussian noise is generated using a normal distribution with:

- Mean = 0
- Standard deviation = 25

The noise is added to the pixel values and the resulting values are clipped to the valid 0–255 intensity range.

This creates a controlled degraded version of each original image.

5.4 Median Filtering

A **5 × 5 Median Filter** is applied to the degraded images.

Median filtering replaces a pixel value with the median value of its neighbourhood. It is commonly used for noise reduction while attempting to preserve image edges.

The output of this stage is compared with the original image to determine the amount of improvement.

5.5 Gaussian Filtering

A **5 × 5 Gaussian Filter** is also applied to the degraded images.

Gaussian filtering performs smoothing using a weighted neighbourhood. It reduces high-frequency noise and produces a smoother image.

The Gaussian-filtered output is evaluated against the original image.

5.6 Contrast Enhancement

After Median Filtering, CLAHE is applied to improve local contrast.

CLAHE stands for **Contrast Limited Adaptive Histogram Equalization**.

The image is converted to the LAB colour space and CLAHE is applied to the L channel, which represents lightness. The image is then converted back to RGB.

This stage is mainly intended to improve the visual appearance and visibility of image details.

5.7 Evaluation

Three metrics are used:

Mean Squared Error (MSE)

MSE measures the average squared difference between the original and processed image.

A lower MSE indicates lower reconstruction error.

Peak Signal-to-Noise Ratio (PSNR)

PSNR measures the quality of the restored image compared with the original image.

A higher PSNR generally indicates better reconstruction quality.

Structural Similarity Index (SSIM)

SSIM measures structural similarity between the original and restored images.

An SSIM value closer to 1 indicates greater structural similarity.

6. IMPLEMENTATION

6.1 Programming Environment

The project was implemented using:

- **Programming Language:** Python
- **Platform:** Google Colab
- **Image Processing Library:** OpenCV
- **Numerical Processing:** NumPy
- **Visualization:** Matplotlib
- **Evaluation:** scikit-image

6.2 Implementation Process

The implementation was performed in the following sequence:

Four Original Images

↓

Image Loading

↓

Resize to 512×512

↓

Add Gaussian Noise

↓

Generate Degraded Images

↓

Median Filtering



Gaussian Filtering



CLAHE Contrast Enhancement



Visual Comparison



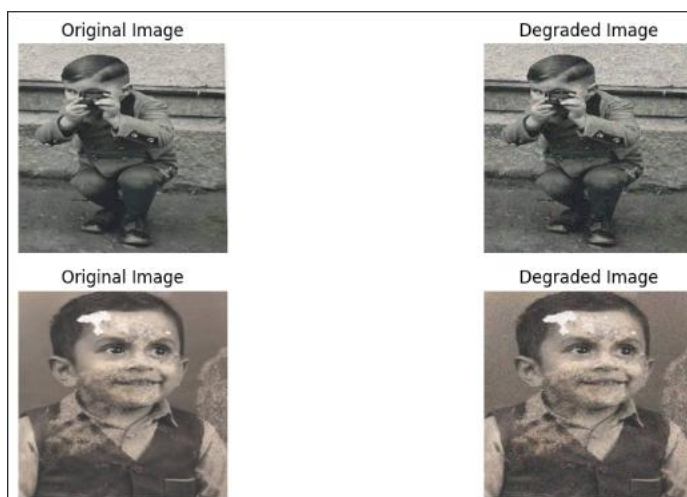
MSE / PSNR / SSIM



Performance Comparison

6.3 Image Degradation Result

Gaussian noise was added to the four original images. The resulting images showed visible degradation compared with the original images.



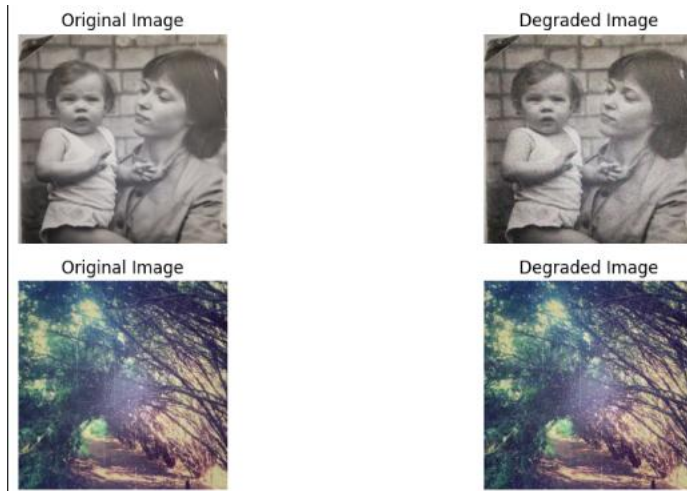


Figure 2: Comparison between original and Gaussian-noise degraded images

6.4 Restoration Results

The restoration techniques were applied to the degraded images.

For visual comparison, the following outputs were generated:

- Original image
- Degraded image
- Median Filter output
- Gaussian Filter output
- Enhanced Restoration output

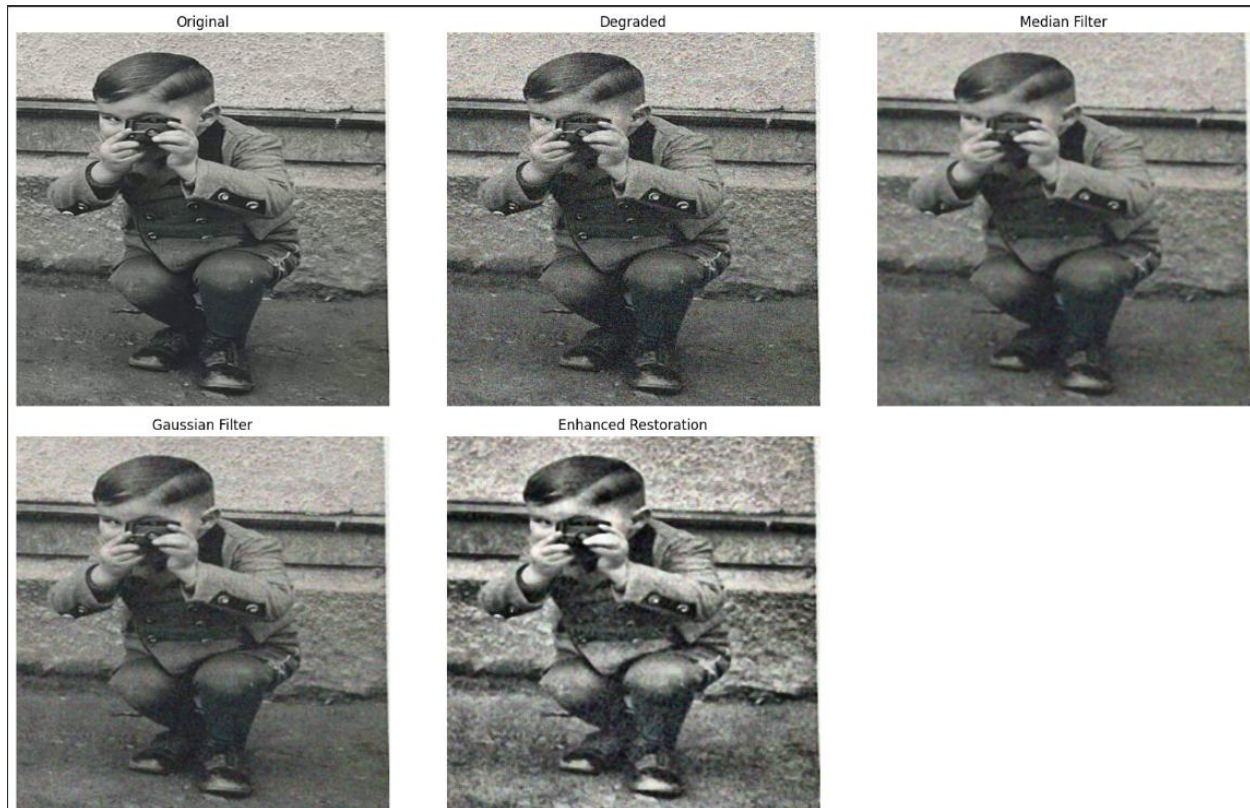


Figure 3: Comparison of restoration techniques

7. RESULTS AND DISCUSSION

The restoration methods were evaluated using the four images.

The average results obtained from the experiment are shown below.

Method	Average MSE	Average PSNR	Average SSIM
Degraded Image	618.04	20.22 dB	0.501
Median Filter	137.44	27.56 dB	0.750
Gaussian Filter	93.77	28.76 dB	0.833
Enhanced Restoration	696.63	19.78 dB	0.618

	Method	Average MSE	Average PSNR	Average SSIM
0	Degraded	618.035428	20.220705	0.500846
1	Median Filter	137.438083	27.563390	0.750080
2	Gaussian Filter	93.773228	28.761554	0.832694
3	Enhanced Restoration	696.626647	19.778454	0.618456

Figure 4: Quantitative evaluation results for the four images

7.1 Analysis of the Degraded Images

The degraded images had an average MSE of **618.04**, PSNR of **20.22 dB**, and SSIM of **0.501**.

The relatively high MSE and lower PSNR/SSIM values indicate that the addition of Gaussian noise caused a noticeable difference from the original images.

The SSIM value of 0.501 also indicates that the structural similarity was reduced after degradation.

7.2 Analysis of Median Filtering

After applying Median Filtering, the average MSE decreased from **618.04 to 137.44**.

At the same time, PSNR increased from **20.22 dB to 27.56 dB**, while SSIM increased from **0.501 to 0.750**.

This indicates that Median Filtering successfully reduced a significant portion of the degradation and restored considerable similarity to the original images.

However, its results were not as strong as Gaussian Filtering for the particular images used in this experiment.

7.3 Analysis of Gaussian Filtering

Gaussian Filtering produced the best quantitative results among the tested restoration methods.

The average values were:

- **MSE = 93.77**
- **PSNR = 28.76 dB**
- **SSIM = 0.833**

Compared with the degraded images, the MSE was substantially reduced, while both PSNR and SSIM increased.

The SSIM value of **0.833** indicates that the Gaussian-filtered images retained a high level of structural similarity with the original images.

Therefore, for the four images and the Gaussian noise level used in this experiment, Gaussian Filtering provided the strongest quantitative restoration performance.

7.4 Analysis of Contrast Enhancement

CLAHE was applied after Median Filtering to improve local contrast.

Although this processing can improve the visual appearance of an image, the quantitative results were lower than those of the filtering methods.

The enhanced restoration produced:

- **MSE = 696.63**
- **PSNR = 19.78 dB**
- **SSIM = 0.618**

The higher MSE and lower PSNR indicate that the enhanced image had greater pixel-level differences from the original reference image.

This does not necessarily mean that the image looked worse visually. CLAHE intentionally modifies local intensity values to improve contrast. Therefore, the enhanced image can appear more visually detailed while still producing a lower numerical similarity to the original image.

7.5 Overall Comparison

The comparison shows that different image-processing techniques serve different purposes.

Median Filtering and Gaussian Filtering primarily focus on reducing noise, whereas CLAHE focuses on improving contrast.

For the particular four images used in this project:

Gaussian Filtering produced the best overall quantitative restoration result.

It achieved:

- The **lowest average MSE: 93.77**
- The **highest average PSNR: 28.76 dB**
- The **highest average SSIM: 0.833**

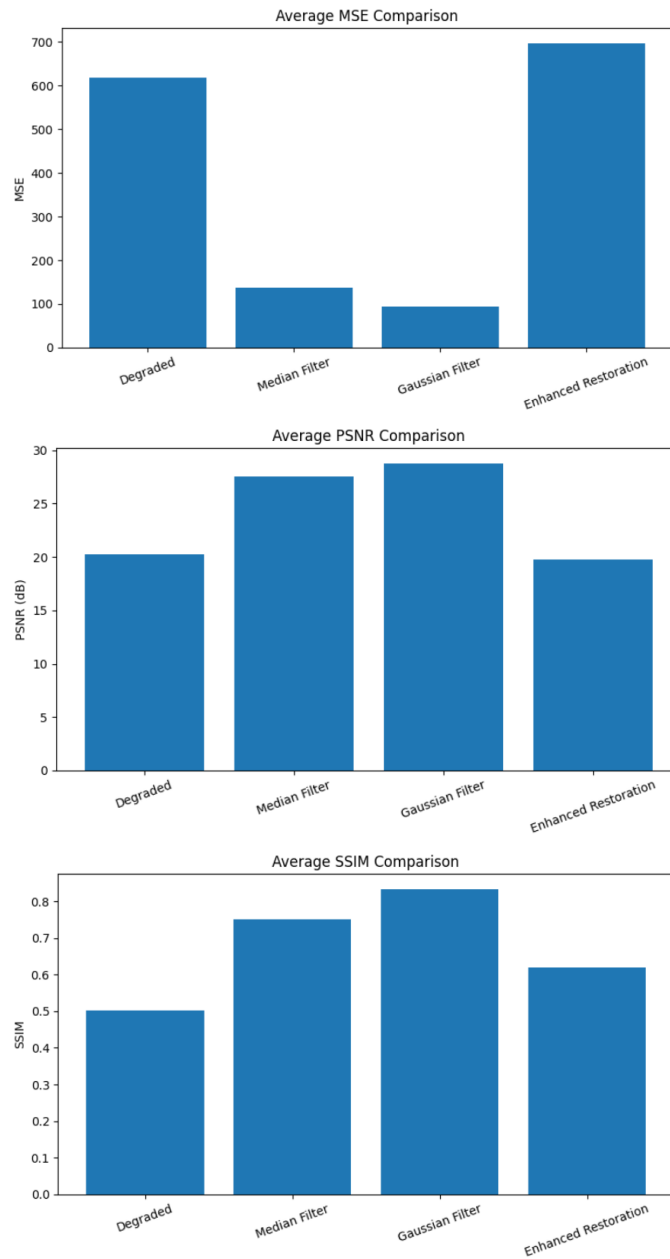


Figure 5: Graphical comparison of MSE, PSNR, and SSIM

8. LIMITATIONS

The project has the following limitations:

1. Only four images were used for experimentation.
2. The images were artificially degraded using Gaussian noise rather than being collected from genuinely damaged old photographs.

3. Only a limited number of restoration techniques were tested.
4. The selected filter parameters, such as the 5×5 kernel, can affect the final result.
5. Filtering may remove some fine details along with unwanted noise.
6. CLAHE changes image intensity values, which can reduce numerical similarity even when the visual contrast improves.
7. The results may differ for other types of degradation such as motion blur, compression artifacts, scratches, or uneven illumination.

9. FUTURE SCOPE

The project can be extended in several ways.

Larger Dataset

A larger collection of real degraded photographs can be used to obtain more general results.

Real Old Photographs

Instead of artificially adding Gaussian noise, actual old photographs containing scratches, blur, fading, and other degradation can be processed.

Additional Noise Models

Other degradation types such as salt-and-pepper noise, speckle noise, and different blur models can be tested.

Advanced Restoration

More advanced restoration and deblurring techniques can be investigated for heavily degraded images.

Deep Learning

Deep learning-based image restoration methods could be used for automatic restoration of complex real-world degradation.

Application Development

The restoration system could be converted into a web or desktop application where users upload a damaged photograph and receive a restored version.

10. CONCLUSION

This project demonstrated the application of Digital Image Processing techniques for restoring degraded images.

Four images were used as reference images, and Gaussian noise was artificially introduced to simulate image degradation. Median Filtering, Gaussian Filtering, and CLAHE-based contrast enhancement were then applied to improve the degraded images.

The experimental results showed clear improvement after the filtering operations. Median Filtering reduced the average MSE from **618.04 to 137.44**, while Gaussian Filtering reduced it further to **93.77**. Gaussian Filtering also achieved the highest average PSNR of **28.76 dB** and SSIM of **0.833** among the tested methods.

CLAHE improved local contrast but produced lower quantitative similarity because contrast enhancement changes the intensity distribution of the image.

Based on the four images used and the Gaussian noise level selected in this experiment, **Gaussian Filtering provided the best quantitative restoration performance**. The project demonstrated that image restoration can be evaluated using both visual inspection and objective measures such as MSE, PSNR, and SSIM.

Overall, the project provided practical understanding of image degradation, spatial filtering, contrast enhancement, and quantitative evaluation of image restoration techniques.

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